

SCENE-LEVEL TYPE-CONSISTENT RELABELLING FOR TRANSPARENT GLASS DETECTION

Heunseung Lim*, Jinhoi Kim, Youngmin Son, Jihyeon Ryu, Seongwoong Kim, and Gichul Yoo*

SK intellix

ABSTRACT

GlassNICOL, a transparent-glass detection dataset, types each glass automatically from a depth-derived height in every view. Type is constant, so adjacent views should agree; the released training labels often disagree, while the hand-labelled test splits do not. We repair the labels with the structure that produced them. The scenes are static and only the camera moves, so every view of a glass is a view of one point on the table: we register views through the table plane, carry each box by its base point, chain the views of each glass, and give every box in a chain the chain’s most frequent type. Detectors trained on the relabelled set, five (accelerator, seed) cells under a sealed protocol, gain out of distribution in type accuracy and six-class average precision over raw labels and over an adjacent-frame vote; on the standard split they gain nothing and lose about two points of recall. Only type ids change.

Index Terms— transparent object detection, label noise, multi-view consistency, relabelling, glass detection

1. INTRODUCTION

Transparent objects are expensive to label. Depth sensors return little through glass and boundaries are weak, so ground truth usually comes from a proxy the sensor can see. ClearGrasp swaps each object for a spray-painted copy [1]; KeyPose labels keypoints in a few views and propagates them through known camera poses, using an opaque twin only for depth [2]; GlassNICOL photographs each tabletop scene in 25 views, records each scene again with the glasses capped to recover depth, and assigns each box a glass type by matching its depth-derived height to the nearest known glass [3]. KeyPose checks its geometry across views; nothing checks that the type assigned in one view agrees with the next.

We check. After registering adjacent training views and matching boxes, 18.1% of matched box pairs that are the same physical glass carry a different type; the hand-labelled test splits pass the same test at 0.0% and 0.3% (Section 2.1).

The structure that produced the labels can also repair them, and the scene says how. A GlassNICOL scene is static: the glasses stand on the table and only the head camera moves. Every view of a glass is therefore a view of one point of the table plane, its base, and a homography of that plane carries points of it exactly from one view to another. We register views pairwise through the table plane, carry each box by its base point, link boxes whose base points coincide, and merge the links into one chain per glass across the scene; each chain gives its most frequent type to every box in it (Fig. 1, Section 2). Our earlier baseline linked only adjacent frames by box overlap after a partial-affine warp, and a missed frame, a failed registration, or the parallax of a tall glass breaks such a chain, so its boxes get a median

of 5.5 votes. Registering through the table plane and carrying base points over two hops gives a median of 10 votes, and on the manual splits the chains merge no two glasses of different type. 12.3% of type ids change and no box moves.

Contributions: (i) a measurement of type-label inconsistency in a released benchmark against its own manual test labels; (ii) a scene-level relabelling method that uses only the dataset’s own views, associates through the support plane without using camera poses or depth, and changes type ids alone; (iii) evidence over five (accelerator, seed) cells on two accelerators, under a protocol fixed before training, against both the raw labels and the tracklet vote (Section 3).

Relation to prior work. Repeat-factor sampling [4] and data pruning [5, 6] choose which images the network sees and take every label as given; relabelling changes what a label says. Label noise is usually found from model confidence [7, 8] or the training loss [9], or absorbed by robust training [10], in detectors too [11]; we locate it from the capture geometry, without a model. Majority voting along tubelets or tracklets smooths a detector’s output at test time [12, 13, 14, 15], and offboard auto-labelling aggregates detections over a sequence into training labels [16]; we vote on a dataset’s released labels, not on detections. Ground-plane homographies associate people across cameras by their feet [17]; we use the table plane the same way, across the views of one moving camera. Label Fusion reprojects one 3D label into every frame [18]; GlassNICOL types each view separately, which is where the disagreement enters. KeyPose discards scans whose keypoints disagree [2]; we keep the frame and correct the label.

2. PROPOSED METHOD

2.1. Type labels disagree on the same object

GlassNICOL assigns type automatically (we call this step the labeller): the height of a depth cluster above the table plane, measured in the capped pass, is matched to the nearest of six known glass heights (5, 9, 11, 13, 17.5 and 22 cm) within 3 cm; the heights and that tolerance are constants of the released labelling code rather than of the paper [3]. Height is measured per view, but type is constant for the object, so one glass should carry one type across the 25 views of its scene. We take the 2,280 consecutive frame pairs of the training scenes, register frame f to $f+1$ by background features, carry each box through the registration and match it to the box it overlaps in $f+1$ at intersection over union (IoU) above 0.5; pairs whose registration fails are matched unwarp. Of the 2,775 matched pairs, 502 (18.1%) carry a different type (Fig. 2), 87% of them between glasses of neighbouring height.

The hand-labelled official splits pass the same registered test: 0 flips in 93 matched pairs on the standard test, at most 1 in 316 (0.3%) on the out-of-distribution (OOD) test. The inconsistency is the labeller’s, not the scenes’. The same labeller also places two

*Corresponding authors: Heunseung Lim and Gichul Yoo.

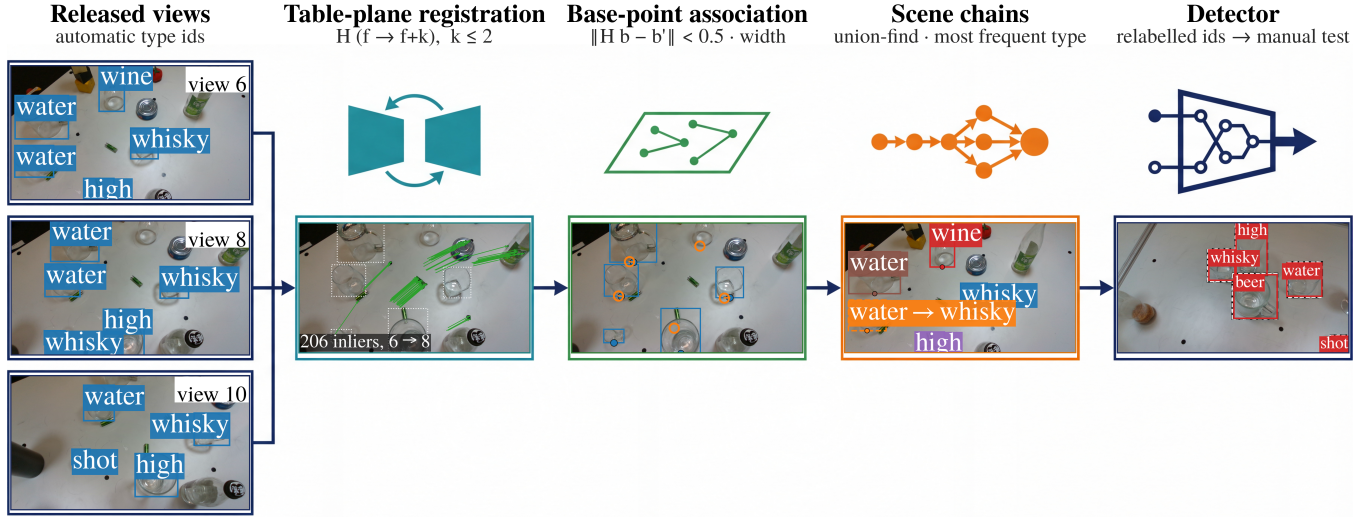


Fig. 1. Overview. Views of a static scene are registered through the table plane by background features; each box is carried by its base point, which lies on that plane, and linked to the box it lands on; union-find merges the links into one chain per glass, whose most frequent type replaces every type id in it; the relabelled set trains a detector scored on the hand-labelled splits. Photographs: views 6, 8 and 10 of one training scene with released boxes and types; inliers of the view 6 to 8 homography as displacement vectors; base points of view 6 carried into view 8 (open) and linked to those of view 8 (filled); view 6 with one colour per chain and a dashed box where the vote replaced the type; a scene-vote detector’s boxes and types on a manual OOD test frame, manual boxes dashed; frame chosen where every manual box is detected with the correct type and nothing else; the pairs the vote does not fix are in Table 5. Photographs and overlays are real; the stage symbols are schematic.

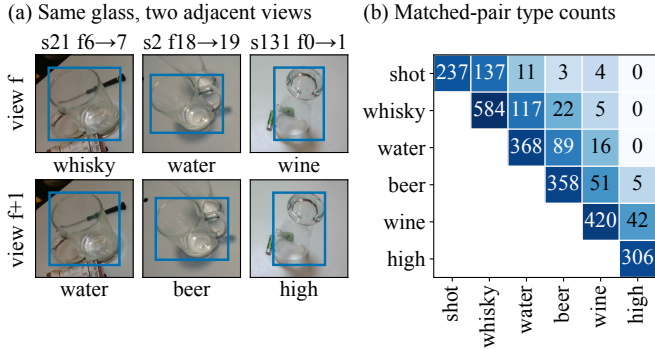


Fig. 2. Automatic type labels are not constant on one object. (a) The same glass in adjacent training views f and $f+1$, cropped from released frames with the released box and type. (b) Released types of the 2,775 box pairs matched across adjacent views after background registration, each unordered pair counted once; 502 (18.1%) disagree, 87% of them between neighbouring heights. Manual test labels flip 0/93 and 1/316.

boxes on one glass (IoU above 0.5) in 51 of 2,375 training frames (44 of 53 such pairs with different types); the manual splits contain none.

The misreads mostly join neighbouring heights and mostly run downward. Treating the views of one glass on an adjacent-frame chain (the tracklets of Section 2.3) as repeated readings and estimating the labeller’s confusion by expectation–maximisation [19], a water glass is read as a shorter type in 18.5% of views and as a taller one in 4.7%, a beer glass 15.9% against 0.9%, a wine glass 11.0% against 0.8%, a highball 11.9%; only the shot glass, which has no shorter type, is misread upward, as whisky in 8.9% (Table 1).

Table 1. The labeller’s confusion, estimated from adjacent-frame view chains by expectation–maximisation: percentage of views in which a glass of the true type (column) is read as each type (row). Misreads of every type but the shortest mostly go toward shorter glasses.

Read as	shot	whisky	water	beer	wine	high
shot	91.1	9.4	4.0	0.4	0.7	–
whisky	8.9	87.5	14.5	3.0	1.2	–
water	–	2.3	76.8	12.5	1.3	–
beer	–	0.8	2.6	83.2	7.7	1.2
wine	–	–	2.1	0.7	88.3	10.7
high	–	–	–	0.2	0.8	88.1

2.2. Association through the table plane

Frames f and $f+k$ of a scene, $k \leq 2$, are registered by background features: up to 1,500 ORB (oriented FAST and rotated BRIEF) features [20] are extracted outside the label boxes, which are dilated $1.2\times$ and masked out, matched under a 0.75 ratio test, and a homography is fitted by RANSAC (random sample consensus) [21] at a 3-pixel threshold and kept when at least 30 matches and half of them are inliers. The background is the table, so the homography is that plane’s and carries points lying on it exactly. We take a glass’s base point to be the bottom-centre of its box, an approximation the tolerance below absorbs; the rest of the box moves with parallax. The release annotates a base point per glass for the training split only, not for the hand-labelled splits where we check the association, so we read it off the box. Each box of f is carried into $f+k$ by its base point and linked, one-to-one in order of distance, to the box whose base point lies within 0.5 of the mean box width. Union-find merges links across the scene into chains, one glass seen in several views, under one constraint: a chain holds at most one box per frame, and a

Table 2. Association ablation on the training split: the first row is the earlier partial-affine, adjacent-frame baseline; the other rows use the homography registration and vary the linking cue (warped-box overlap or base point) and the hop count. Chains; median votes per box; boxes with three or more votes (%); type ids changed (%); manual-split checks on the standard (Std.) and OOD splits (first row: flips among matched adjacent-view pairs, Section 2.1; other rows: disagreeing boxes).

Association	Chains	Votes	≥ 3	Chg.	Std.	OOD
adjacent overlap (tracklet)	1,561	5.5	71	8.4	0/93	1/316
overlap, 1 hop	1,115	8	82	9.2	0/167	0/351
overlap, 2 hops	1,012	9	85	10.3	0/167	0/351
base point, 1 hop	893	9	88	10.9	0/167	2/351
base point, 2 hops (ours)	786	10	91	12.3	0/167	0/351

link that would violate it is refused (64 of the links proposed by the 4,097 registered pairs).

2.3. Vote, coverage and validity

Each chain takes its most frequent type (a plurality); ties are left unchanged. Only the type id changes: no box moves, no image is added or removed, and roster and hyperparameters are identical to the raw-label arm of Section 3. The association runs on all 95 training scenes (the detector roster below uses 70 of them, on which 11.7% of type ids change; the other 25 stay out of training): 4,097 of 4,465 frame pairs register; 4,336 boxes fall into 786 chains with a median of 10 views per box, 3,953 boxes (91%) are voted on by three views or more, 535 boxes (12.3%) change type and 140 in tied chains keep their label. Table 2 separates the ingredients: the adjacent-frame overlap vote of our earlier baseline, which registered 1,748 of 2,280 pairs by a partial-affine warp, gives 1,561 chains, 878 of them a single box, a median of 5.5 votes per box, and changes 364 boxes (8.4%). Under the homography registration, overlap linking over one hop already lengthens the chains; the base point and the second hop each add votes, and only their combination gives the most votes with no merged glasses of different type on either manual split. That baseline is the tracklet arm of Section 3. The validation split is left as released: checkpoint selection uses generic average precision (AP), so type ids never enter it.

Validity. Manual types are constant per object, so on the hand-labelled splits every within-chain type disagreement is a merged pair of glasses. The association above produces 0 of 167 disagreeing boxes on the standard split and 0 of 351 on the OOD split, with median chain lengths of 10 and 15 (163 of 167 and 346 of 351 boxes in chains of three or more). The hop count and tolerance were chosen by this check on a grid of two to four hops and four tolerances before any scene-vote detector was trained, taking the loosest setting that disagrees nowhere on either split, so the zero above is an in-sample value: at the chosen tolerance, three and four hops leave four disagreements on the standard split and two hops none; a one-hop run made afterwards leaves two on the OOD split. The manual splits never enter training.

Scope. A vote over views removes disagreement that varies by view and inherits whatever is constant across views, such as the class-share difference of Section 4; Section 3.3 scores each pair.

3. EXPERIMENTAL RESULTS

3.1. Setup

Data. The official GlassNICOL head-camera split [3]: 2,375 training images from 95 scenes, 125 validation images, and hand-labelled standard and OOD tests (66 and 75 images, 3 scenes each). The training roster, fixed before this experiment, uses 70 of the 95 scenes (1,750 images).

Detector and regime. YOLO11n [22] (2.59M parameters [23]), 640×640 inputs, batch 64, AdamW [24] at $\text{lr}_0 = 10^{-3}$, cosine decay over 300 epochs. All arms train under the same repeated-exposure regime: the 1,538 training images whose transparent and chalk-coated captures register by the same background matching appear twice, 3,288 slots and 15,600 optimiser updates, and the arms differ only in training type ids: raw, tracklet vote, scene vote. Seeds 1337, 3407 and 4567 run on accelerator A (an H200 GPU) and seeds 1337 and 5678 on accelerator B (an RTX PRO 6000 GPU); a cell is one (accelerator, seed) set of the three arms, an arm being one label variant on the shared roster. On A the raw and scene-vote runs of a cell share a GPU and the tracklet run used another; on B all share one.

Scoring. The checkpoint with the best validation generic AP is scored at confidence floor 0.001 with non-maximum suppression (NMS) at IoU 0.7: COCO (Common Objects in Context) box AP over IoU 0.50–0.95 [25]; generic AP collapses the six types to one. Type accuracy is the fraction of ground-truth boxes localised by the compared arms ($\text{IoU} \geq 0.5$, confidence ≥ 0.25) with the correct type; recall is the localised fraction per arm.

Protocol. The comparison was preregistered before any scene-vote detector was trained: the three arms, the five cells, the checkpoint rule, the four metrics above, and four predictions, a positive mean type-accuracy gain over the raw labels on both splits (P1), an OOD gain exceeding the tracklet vote’s (P2), recall and generic AP within two points (P3), and a positive OOD gain in at least 4 of 5 cells (P4). The sealed protocol, every evaluation receipt and the code are at <https://github.com/HeunSeungLim/glassnicol-scene-relabel>. Every contrast is read as a mean over cells with the count of positive cells beside it [26].

3.2. What the labels change

Table 3 pools the seeds per accelerator and Table 4 lists every cell. Two contrasts hold across cells. Out of distribution, six-class AP rises by +7.1 points over the raw labels in 5 of 5 cells (across-cell 95% t -interval [+1.8, +12.3]) and by +6.4 over the tracklet vote ([+0.6, +12.2]), which the tracklet vote itself did not move (+0.7). On the standard split localisation recall falls in every cell, by 2.0 points on average ([−3.7, −0.3]). The type-accuracy gains are larger but less certain: out of distribution the scene vote gains +9.2 points over the raw labels, positive in 4 of 5 cells, with per-cell intervals, resampling test images, above zero in 3 cells, reaching it in 1 and below it in 1 (Table 4); resampling whole scenes instead, of which each split has 3, widens them until 8 of 10 include zero and an across-cell interval of [−5.7, +24.1] (cell standard deviation 12.0); over the tracklet vote it gains +4.0 (4 of 5). On the standard split type accuracy changes by +0.8 over the raw labels and −1.8 over the tracklet vote, with 1 cell above zero and 1 below, and six-class and generic AP change by −1.9 and −1.7 (2 and 1 of 5 cells positive). The one cell that loses on both splits is seed 4567 on A, by eight and ten points; every other gains out of distribution by six or more.

Preregistered predictions. P1 held: the mean type-accuracy gain over the raw labels is positive on both splits, +0.8 and +9.2. P2 held:

Table 3. Raw, tracklet-vote and scene-vote training labels on the official test splits (manual labels), mean $\pm$ standard deviation over seeds per accelerator A or B (number of seeds in parentheses; with two seeds the deviation has one degree of freedom). Six: six-class AP; Gen: generic AP; Rec: localisation recall; Type: type accuracy on boxes localised by all three arms, a different box set from the pairwise contrasts of Table 4; all in percent.

Accel. (seeds)	Labels	Standard				OOD			
		Six	Gen	Rec	Type	Six	Gen	Rec	Type
A (3)	raw	68.9 $\pm$ 3.9	84.0 $\pm$ 2.2	89.6 $\pm$ 1.2	86.1 $\pm$ 4.2	53.8 $\pm$ 7.1	66.1 $\pm$ 0.9	77.9 $\pm$ 4.1	79.9 $\pm$ 7.3
	tracklet vote	64.6 $\pm$ 4.6	84.8 $\pm$ 3.0	88.6 $\pm$ 1.2	86.3 $\pm$ 3.0	54.0 $\pm$ 4.6	67.3 $\pm$ 1.0	81.0 $\pm$ 0.8	82.6 $\pm$ 2.3
	scene vote	61.9 $\pm$ 2.7	81.6 $\pm$ 3.5	87.6 $\pm$ 1.2	82.6 $\pm$ 2.0	60.2 $\pm$ 3.2	65.9 $\pm$ 1.4	77.1 $\pm$ 4.1	83.8 $\pm$ 9.4
B (2)	raw	60.1 $\pm$ 1.6	84.4 $\pm$ 2.2	92.5 $\pm$ 1.3	77.5 $\pm$ 2.8	53.5 $\pm$ 0.4	67.4 $\pm$ 1.5	82.3 $\pm$ 2.0	68.6 $\pm$ 5.6
	tracklet vote	67.0 $\pm$ 2.8	87.2 $\pm$ 0.7	89.5 $\pm$ 3.8	83.9 $\pm$ 0.7	54.8 $\pm$ 7.4	66.9 $\pm$ 2.9	78.8 $\pm$ 1.4	76.4 $\pm$ 7.9
	scene vote	65.9 $\pm$ 2.6	83.8 $\pm$ 2.5	90.4 $\pm$ 3.4	84.3 $\pm$ 5.2	61.5 $\pm$ 1.5	66.8 $\pm$ 2.3	80.3 $\pm$ 1.6	84.9 $\pm$ 0.1

Table 4. Type-accuracy gain of the scene vote in every (accelerator, seed) cell, in points on the manual boxes both compared arms localise: over the raw labels; 95% paired frame-bootstrap interval (2,000 draws).

Accel.	Seed	Standard		OOD	
		Gain	95% interval	Gain	95% interval
A	1337	-5.6	[-11.0, +0.0]	+6.2	[-0.4, +13.1]
	3407	+4.2	[-4.1, +11.8]	+15.9	[+10.4, +22.0]
	4567	-8.0	[-14.9, -1.5]	-10.4	[-15.2, -6.0]
B	1337	+8.5	[+2.2, +15.1]	+14.4	[+10.4, +19.0]
	5678	+4.7	[-3.4, +13.2]	+19.9	[+14.6, +24.9]

Table 5. Type-accuracy change of the scene vote over the raw labels on the manual test boxes of each confusable pair (localised by both arms): mean over the 5 cells and the number of positive cells.

Pair	Standard		OOD	
	Mean	Positive	Mean	Positive
shot/whisky	-4.1	1/5	+8.2	3/5
whisky/water	-6.9	1/5	+14.9	4/5
water/beer	+3.2	2/5	+8.0	4/5
wine/high	+3.8	4/5	+11.1	4/5

out of distribution the scene vote’s gain exceeds the tracklet vote’s gain of +5.0. P4 held: 4 of 5 OOD cells are positive. P3 did not hold: on the standard split localisation recall falls by 2.0 points, just past the two-point bound, and generic AP by 1.7; out of distribution both stay within it. Scoring the last epoch instead (post hoc, rows released with the code) gives, over the raw labels, type accuracy -0.6 on the standard split (2 of 5 cells, so P1 holds only under the preregistered rule) and +11.3 out of distribution (5 of 5), six-class AP -1.7 and +11.1, recall +0.5 and -1.7 (2 and 2 of 5 cells positive). The OOD gain holds under both rules; the standard-split recall loss does not. Under that rule the tracklet vote also gains out of distribution, six-class AP +7.0 in 5 of 5 cells, so the separation between the two votes belongs to the preregistered checkpoint rule.

3.3. Where the vote helps and where it cannot

Table 5 scores each confusable pair against the raw labels. Out of distribution every pair gains on average: whisky/water +14.9 points (4/5 cells), wine/high +11.1 (4/5), water/beer +8.0 (4/5) and shot/whisky

+8.2 on average (3/5), a mean carried by one cell at +52.6 with a median of +1.8; that is the pair whose share differs most between training and test and which the tracklet vote could not move. On the standard split the two shortest pairs lose, shot/whisky -4.1 and whisky/water -6.9, while water/beer and wine/high gain +3.2 and +3.8.

4. DISCUSSION AND LIMITATIONS

The vote does not bring class shares closer to the tests. Whisky is 25.9% of raw training boxes, 17.4% of the standard test and 11.7% of the OOD test, and the shot:whisky ratio is 0.58 in raw training labels, 0.51 after the scene vote, against 0.79 and 1.83 in the tests. Decoding the chains under the confusion of Table 1 instead of by the most frequent type moves the ratio no closer; deciding whether the difference is the labeller’s or the staging’s needs the per-box height the labeller measured, which is not released; we did not recompute it from the released depth, nor run a pose-based reprojection baseline.

The association is only as good as the table: frames without enough background features stay unlinked, a chain cannot vote on a glass the labeller never boxed, and the 51 frames with two boxes on one glass are voted on as two objects.

The gain is out of distribution; the standard-split costs are in Section 3. One seed on A loses on both splits, and with five cells the OOD type-accuracy mean sits inside the seed spread, so the six-class AP contrast carries the claim. The official test splits have three scenes each, and the tracklet contrast re-run for seed 1337 on accelerator B moved by 10.9 and 14.3 type-accuracy points, so single cells are read against that swing. Checkpoint selection by validation generic AP ($r^2 = 0.04$ with test six-class AP) is the same for every arm, so the arms differ only in their labels.

5. CONCLUSION

GlassNICOL’s automatic type ids disagree on the same glass in 18.1% of adjacent-view pairs; its hand labels do not. The table plane links the views of a glass, and a vote over them gives one type per object: 12.3% of type ids change and no box moves. Detectors trained on the corrected labels gain out of distribution, six-class AP by +7.1 points over the raw labels in every cell and +6.4 over an adjacent-frame vote; the type-accuracy gains are larger but straddle zero across cells. On the standard split they gain nothing and lose two points of recall.

6. REFERENCES

- [1] Shreeyak S. Sajjan, Matthew Moore, Mike Pan, Ganesh Nagaraja, Johnny Lee, Andy Zeng, and Shuran Song, “ClearGrasp: 3d shape estimation of transparent objects for manipulation,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2020, pp. 3634–3642.
- [2] Xingyu Liu, Rico Jonschkowski, Anelia Angelova, and Kurt Konolige, “KeyPose: Multi-view 3d labeling and keypoint estimation for transparent objects,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 11599–11607.
- [3] Lukáš Gajdošech, Hassan Ali, Jan-Gerrit Habekost, Martin Madaras, Matthias Kerzel, and Stefan Wermter, “Shaken, not stirred: A novel dataset for visual understanding of glasses in human-robot bartending tasks,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2025, pp. 20516–20523.
- [4] Agrim Gupta, Piotr Dollár, and Ross Girshick, “LVIS: A dataset for large vocabulary instance segmentation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 5356–5364.
- [5] Mansheej Paul, Surya Ganguli, and Gintare Karolina Dziugaite, “Deep learning on a data diet: Finding important examples early in training,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.
- [6] Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos, “Beyond neural scaling laws: Beating power law scaling via data pruning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [7] Curtis G. Northcutt, Lu Jiang, and Isaac L. Chuang, “Confident learning: Estimating uncertainty in dataset labels,” *J. Artif. Intell. Res.*, vol. 70, pp. 1373–1411, 2021.
- [8] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller, “Pervasive label errors in test sets destabilize machine learning benchmarks,” in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, 2021.
- [9] Marius Schubert, Tobias Riedlinger, Karsten Kahl, Daniel Kröll, Sebastian Schöner, Siniša Šegvić, and Matthias Rottmann, “Identifying label errors in object detection datasets by loss inspection,” in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2024, pp. 4582–4591.
- [10] Hwanjun Song, Minseok Kim, Dongmin Park, Yooju Shin, and Jae-Gil Lee, “Learning from noisy labels with deep neural networks: A survey,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8135–8153, 2023.
- [11] Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, and Yixuan Yuan, “Towards robust adaptive object detection under noisy annotations,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 14187–14196.
- [12] Kai Kang, Wanli Ouyang, Hongsheng Li, and Xiaogang Wang, “Object detection from video tubelets with convolutional neural networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 817–825.
- [13] Wei Han, Pooya Khorrami, Tom Le Paine, Prajit Ramachandran, Mohammad Babaeizadeh, Honghui Shi, Jianan Li, Shuicheng Yan, and Thomas S. Huang, “Seq-NMS for video object detection,” *arXiv preprint arXiv:1602.08465*, 2016.
- [14] Subarna Tripathi, Serge Belongie, Youngbae Hwang, and Truong Nguyen, “Detecting temporally consistent objects in videos through object class label propagation,” in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2016, pp. 1–9.
- [15] Yang Chen, Changsoo S. Jeong, Deepak Khosla, Kyungnam Kim, Shinko Y. Cheng, Lei Zhang, and Alexander L. Honda, “Object recognition consistency improvement using a pseudo-tracklet approach,” U.S. Patent 9,342,759 B1, HRL Laboratories, May 2016.
- [16] Charles R. Qi, Yin Zhou, Mahyar Najibi, Pei Sun, Khoa Vo, Boyang Deng, and Dragomir Anguelov, “Offboard 3D object detection from point cloud sequences,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 6134–6144.
- [17] Saad M. Khan and Mubarak Shah, “A multiview approach to tracking people in crowded scenes using a planar homography constraint,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2006, vol. 3954 of *LNCS*, pp. 133–146.
- [18] Pat Marion, Peter R. Florence, Lucas Manuelli, and Russ Tedrake, “Label Fusion: A pipeline for generating ground truth labels for real RGBD data of cluttered scenes,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2018, pp. 3235–3242.
- [19] A. P. Dawid and A. M. Skene, “Maximum likelihood estimation of observer error-rates using the EM algorithm,” *J. R. Stat. Soc. C (Appl. Stat.)*, vol. 28, no. 1, pp. 20–28, 1979.
- [20] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski, “ORB: An efficient alternative to SIFT or SURF,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2011, pp. 2564–2571.
- [21] Martin A. Fischler and Robert C. Bolles, “Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography,” *Commun. ACM*, vol. 24, no. 6, pp. 381–395, 1981.
- [22] Glenn Jocher and Jing Qiu, “Ultralytics YOLO11,” 2024, Software, <https://github.com/ultralytics/ultralytics>.
- [23] Ranjan Sapkota, Marco Flores-Calero, Rizwan Qureshi, Chetan Badgujar, Upesh Nepal, Alwin Poullose, Peter Zeno, Uday Bhanu Prakash Vaddevolu, Sheheryar Khan, Maged Shoman, Hong Yan, and Manoj Karkee, “YOLO advances to its genesis: A decadal and comprehensive review of the you only look once (YOLO) series,” *Artif. Intell. Rev.*, vol. 58, no. 9, 2025, art. no. 274.
- [24] Ilya Loshchilov and Frank Hutter, “Decoupled weight decay regularization,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2019.
- [25] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick, “Microsoft COCO: Common objects in context,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 740–755.
- [26] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent, “Accounting for variance in machine learning benchmarks,” in *Proc. Mach. Learn. Syst. (MLSys)*, 2021, vol. 3, pp. 747–769.